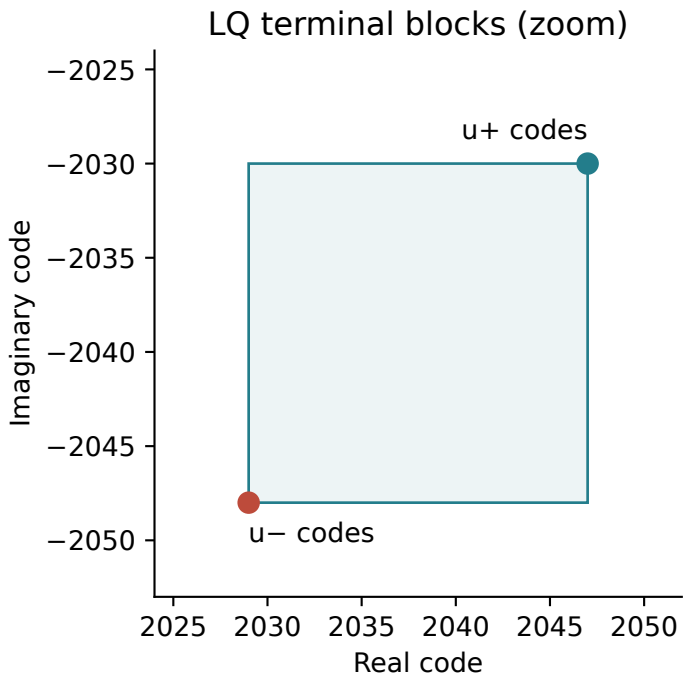
## From every incoming grid state

Real codes: $r' \geq \min(U - 18, r + 149)$
At most 28 pairs to $r \geq U - 18$

Imaginary codes: $i' \leq \max(L + 18, i - 62)$
At most 66 pairs to $i \leq L + 18$

Next pair → exact reversed signature
67 is sufficient; no minimality claim.